Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Vortex

Project / Solution Name

Vortex OrderGuard

Name (Leader)

Aun Yi Qi

Email & Phone (Leader)

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University Affiliation(s)

MMU

Live Prototype URL

https://sdc-hackathon-submission-template.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/Aunyiqi/SDC_Hackathon_Submission_Template

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The prototype accepts synthetic purchasing records through a Flask web interface hosted on Replit, with demonstration cases stored in SQLite. Optional AI-powered extraction uses DeepSeek to process supplier text or a RunPod Serverless GPU worker running EasyOCR to extract information from synthetic invoice evidence. All extracted fields are treated as suggestions and require human verification.

Deterministic Python rules cross-check purchase orders, receiving records and invoices, while the **Trust Gate** blocks ambiguous cases for further review. An authorised owner can clarify, approve or reject each case. Approved actions are recorded in a simulated outbox and activity log for traceability.

The prototype operates entirely in a demonstration environment. It does not send real supplier messages, process payments or connect to live Orbyte APIs. Manual data entry remains available if an external AI service is unavailable.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Flask 3.1.3	Web routes / sessions	BSD-3-Clause
Requests 2.34.2	External HTTP APIs	Apache-2.0
Bootstrap 5.3.3	Responsive UI	MIT
RunPod 1.12.0; EasyOCR 1.7.2; OpenCV ≥4.10, <5; Pillow ≥10, <12; NumPy <1.26.3	GPU OCR worker dependencies	Apache-2.0; MIT; HPND; BSD-3-Clause

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek API	Optional supplier-text extraction	Paid; Usage billed by tokens
RunPod Serverless API	OCR worker invocation	Sponsor-provided

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Web app hosting/deployment	
RunPod Serverless GPU	EasyOCR inference worker	
SQLite	Demo data store	

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek Chat	DeepSeek	DeepSeek API / pre-trained model	DeepSeek's API and provider terms apply;
EasyOCR recognition model	Jaided AI	Pre-trained OCR via EasyOCR 1.7.2	Apache-2.0

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic purchasing cases	-	Synthetic	-	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent	Repository records review, code fixes, regression tests, and documentation/PR support	Moderate

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Replit Agent supported code reviews, bug fixes, regression testing and documentation. Team members reviewed AI-generated changes, verified fixes through testing and checked documentation for accuracy before accepting the work. Each contributor remains responsible for reviewing and validating their own contributions before submission. DeepSeek was configured as an optional supplier-text extraction service, but its use in the deployed demo has not been confirmed.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Credits	Web app hosting/deployment
RunPod	Credits	EasyOCR inference worker

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit and RunPod played important roles in enabling the prototype. Replit supported development, testing and deployment of the Flask application, while RunPod provided serverless GPU infrastructure for OCR-based invoice extraction. Although alternative tools could replace them, they significantly supported the implementation and demonstration of the prototype's intended workflow.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No actual customer, supplier or other user data is used; all demonstration records are synthetic. The Flask application stores demonstration cases in SQLite and sends extraction inputs to DeepSeek and/or RunPod only when the optional services are configured and invoked. Public demonstration pages are read-only, while data modifications require a shared owner code/session and CSRF protection. However, the shared code does not provide individual user authentication or production-level tenant isolation. Data retention and persistence for synthetic records, logs and uploaded images depend on the deployed hosting configuration and require further verification. API keys and private owner codes must remain confidential.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Optional extraction may omit or misread information, and type validation alone cannot guarantee accuracy. Performance depends on external provider and model availability, API limits, network connectivity and GPU startup time. Manual entry remains available as a fallback, and the system does not perform unit conversions. Deterministic validation rules may still produce incorrect results when inputs are inaccurate or records are incomplete.

To reduce risk, extracted data requires human verification, ambiguous cases are blocked for review, and approvals are restricted to the authorised demo owner. Exceptions retain links to their source records, while simulated outcomes and audit logs support traceability. All demonstration data is synthetic. The demo-owner code does not provide production-grade identity management or live Orbyte role-based access control (RBAC). No real messages or payments are processed, and live OCR/model accuracy and user benefits have not yet been measured.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]

25/10/2026